

LEARNING INTENTION

I can explain how light travels and how shadows, reflection and refraction occur.

SUCCESS CRITERIA

- ☐ I know light travels in a straight path
- ☐ I can explain how shadows are formed
- ☐ I can describe reflection and refraction

KEY CONCEPTS

Reflection: light bounces off a surface (e.g. mirror). **Refraction:** light bends when it passes from one material to another (e.g. water). **Shadow:** formed when an opaque object blocks light.

WORKED EXAMPLE

A shadow is long in the morning because the sun is low — light travels at a low angle, making a long blocked area. At midday, the sun is high so shadows are short.

PART A — WARM UP

1. Does light travel in a straight line or a curved line?
2. What is needed to form a shadow? (List three things.)
3. Name TWO natural sources of light and TWO artificial sources.
4. What word describes an object that blocks all light?

PART B — CORE SKILLS

5. Explain the difference between reflection and refraction.
6. A pencil placed in a glass of water looks bent. Explain why using the word refraction.
7. Why does a torch shine a brighter spot on the wall when held close versus far away?
8. Draw a diagram in your working space showing how a shadow is formed.

LIGHT HELPER

Opaque: blocks all light (solid wall). **Transparent:** lets all light through (clear glass). **Translucent:** lets some light through (frosted glass). Only opaque objects form clear, sharp shadows.

WORKING SPACE — SHADOW DIAGRAM

PART C — CHALLENGE

9. Real-World Context: Lifeguards wear bright yellow vests and use mirrors to signal for help. Explain how both the bright colour AND the mirror use the properties of light to help people see the lifeguard from a distance.

10. Reasoning: A student says "a shadow is a dark object that follows you." Write a scientifically correct explanation of what a shadow actually is, and why it changes size throughout the day.

REFLECTION

Today I learned...

I CAN CHECKLIST

- ☐ I know light travels in straight paths
- ☐ I can explain how shadows are formed
- ☐ I can describe reflection and refraction

TEACHER FEEDBACK

Strength: _____

Next Step: _____

PARENT TIP

On a sunny day, trace shadow outlines at different times and compare — great science in action.

TEACHER TIP

Use a torch and objects to experiment with shadow size and position in a darkened room.

ANSWER KEY (FOR PARENT / TEACHER USE)

Q1: Straight line

Q2: A light source, an opaque object, and a surface behind it

Q3: Natural: sun, fire. Artificial: torch, lamp

Q4: Opaque

Q5: Reflection = bounces off; refraction = bends as it passes through different materials

Q6: Light slows and bends as it moves from water to air, making the pencil appear bent (refraction)

Q7: Light spreads the further it travels; closer = more concentrated beam

Q8: Check diagram shows: light source → object blocking → shadow formed behind

Q9: Yellow colour reflects/absorbs light so it's highly visible; mirror reflects sunlight toward rescuers to signal location

Q10: A shadow is an area where light is blocked by an opaque object; it changes size because the angle of the sun changes throughout the day, changing the angle of the light rays